BASBO AYELAZONO

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RESEARCH INTERESTS

• Human-Computer Interaction • Accessible Computing • Assistive Technology • Artificial Intelligence • Human-Centered Computing

EDUCATION

Dakota State University (Bachelor of Science) Expected Graduation – May 2027
GPA: 3.73/4.00
Major: Computer Science
Minors: Bioinformatics, Integrated Biology
President's Academic Honors: Fall 2023 – Present

RESEARCH EXPERIENCE

Research Assistant August 2025 – Present
Madison Cyber Labs Madison, South Dakota
PI – Dr. Justin Blessinger
Research Focus: Human-Computer Interaction, Accessibility & Assistive Technology

- Conducted pre- and post-intervention user interviews across accessibility-focused projects to identify user needs, evaluate usability and user experience, and inform design decisions.
- Applied human-centered and iterative design methods across accessibility-focused research projects, incorporating user feedback into prototype development and refinement.

Project: Cognitive Accessibility & Multisensory Environment Design

- Designed a Snoezelen multisensory environment using CAD, human-centered design frameworks, accessibility standards, and embedded hardware to support users with cognitive and sensory disabilities.

Project: Assistive Hearing & Audio Signal Processing

- Conducted quantitative audio signal analysis using Python and MATLAB to evaluate amplification, latency, and noise reduction across consumer audio devices.

Project: Assistive Biomedical Device Prototyping

- Designed and fabricated wearable assistive device prototypes using iterative CAD and 3D-printing workflows, evaluating form, fit, and usability.

Project: AI-Based Accessibility Tools

- Evaluated AI-based speech enhancement and transcription tools using Python-based workflows to benchmark noise suppression, transcription accuracy, and usability.

Computational Biology Intern March 2026 – Present
Yale School of Medicine Remote
PI – Dr. Ruslan Medzhitov and Dr. Allison Greaney
Project: Investigating Lung Biology through Tissue-Engineered Bioreactor Models

- Analyze single-cell RNA sequencing (scRNA-seq) data in R using Seurat and ggplot2 to characterize gene expression patterns in tissue-engineered lung models.

- Execute an end-to-end scRNA-seq pipeline, including quality control, normalization, dimensionality reduction, cell clustering, and differential expression analysis, to characterize transcriptional differences associated with BCL5 and BCL6 gene expression in lung tissue models.

Biotechnology Research & Development Intern May 2026 – August 2026
Corteva Agriscience – Genome Center of Excellence Johnston, Iowa
PI – Jennifer Klaiber

- Engineered a full-stack Python and Streamlit application to automate genomic probe panel tracking, replacing a manual spreadsheet-based process that reduced processing time by 98%
- Integrated the application with internal genomic databases via REST APIs and implemented JWT-based authentication (AEGIS), enabling secure, real-time data exchange between R&D and bioinformatics systems
- Built a custom email-parsing workaround using the win32com library to extract structured shipment data from Outlook distribution lists, circumventing Microsoft Graph API limitations and eliminating a manual data-entry step in the logistics workflow
- Collaborated with infrastructure engineering on Azure DevOps CI/CD pipeline design and Bicep-based deployment, improving the reliability and maintainability of research software as it moved toward production use
- Developed a machine learning-based image processing agent for the Genome Editing team, automating cell colony counting from microscopy image data to support editing verification and reduce manual review time

Summer Research Intern May 2025 – August 2025
University of Utah Department of Mechanical Engineering Salt Lake City, Utah
Project: High-Throughput Experimentation and Data Analysis Revealing the Intricate Interplay of Oil Spills and Microplastics Across Diverse Conditions
PI – Dr. Samira Shiri

- Selected as one of 25 scholars in a highly competitive Summer Program for Undergraduate Research
- Designed and executed a high-throughput experimental workflow to investigate particle-surface interactions, generating large-scale image-based datasets for quantitative analysis
- Developed an automated image analysis pipeline using ImageJ and MATLAB to quantify dynamic physical phenomena of polyethylene, polystyrene and oils
- Developed structured workflows using C++ and Excel to improve efficiency, transparency, and reporting accuracy

Undergraduate Ecology Researcher August 2025 – March 2026
Dakota State University Department of Arts & Sciences Madison, South Dakota
PI – Dr. Kristel Bakker

- Conducted field-based survey of Great Blue Heron rookeries in South Dakota, quantifying colony metrics and comparing findings against three decades of prior literature (Baker et al. 2015, Naugle et al. 1996, Dowd 1982)
- Analyzed spatial and habitat data to assess landscape-level ecological change across survey sites
- Investigated neonicotinoid contamination in wild bird eggshells, detecting imidacloprid residues across multiple species using mass spectrometry and microscopy

Undergraduate Biology Researcher August 2024 – May 2025
Dakota State University Department of Arts & Sciences Madison, South Dakota
Project: Survey of Prairie Lakes for Microplastics in Eastern South Dakota
PI – Dr. Kristel Bakker

- Conducted field sampling across four prairie lakes (546–6,570 ha), collecting shoreline and water samples across multiple sites to assess microplastic contamination
- Isolated 1–5 mm particles using sieving, saltwater flotation, and vacuum filtration, then identified microplastics by morphology using light microscopy at 80×–350× magnification
- Characterized microplastic contamination across lake sites, detecting microfilaments in four lakes and pellets in one lake, with microfilaments representing the most commonly detected microplastic type
- Presented findings at National Conference on Undergraduate Research (NCUR) 2025

Software Engineering Fellow June 2024 – July 2024
HeadStarter AI Remote

- Engineered Python-based data processing and automation tools to streamline research workflows
- Performed quantitative validation and data quality checks to maintain accuracy and audit readiness
- Delivered reports and visualizations supporting data-driven decision-making
- Collaborated with stakeholders to translate technical findings into process improvements

TECHNICAL SKILLS

Programming: Python, R, MATLAB, C++, SQL
Bioinformatics: Seurat, scRNA-seq analysis, differential expression, transcriptomics
Machine Learning: Image classification/processing, supervised learning
Data Analysis & Visualization: ggplot2, ComplexHeatmap, ImageJ, Excel
Hardware & Prototyping: CAD, 3D printing, embedded systems
Laboratory: Microscopy, field sampling, experimental workflows

LEADERSHIP EXPERIENCE

Editor-in-Chief December 2023 – Present
Trojan Times – Dakota State University Newspaper Madison, South Dakota

- Lead a team of writers, editors, and designers to produce the university's flagship publication
- Designed and conducted an in-depth interview with the Governor of South Dakota, translating complex policy discussions into accessible, fact-checked reporting
- Oversee the editorial process from pitch to publication, enforcing quality and accuracy standards
- Develop content strategy engaging the campus community on science, research, and technology

AWARDS AND HONORS

- Princeton University P3 Scholar - Princeton University, 2026
- Yale University VIBES Scholar - Yale University, 2026
- University of Utah SPUR Scholar - University of Utah, 2025
- Student Research Initiative Award (2×) – Dakota State University, 2024, 2025
- Second Best Poster Presentation – Dakota State University, 2025
- President's Academic Honors List
- Dakota State University Rising Scholar

- Dakota State University Champion Scholar

SELECTED ABSTRACTS & CONFERENCE PRESENTATIONS

Oral Presentations

- **Ayelazono, B**. (2026, April). *Nest Success and Habitat Characteristics of a First-Year Great Blue Heron Rookery.*
  Presented at the International Association for Landscape Ecology – North America (IALE-NA). Athens, GA.
- **Ayelazono, B**. (2025, April). *Survey of Prairie Lakes for Microplastics in Eastern South Dakota.* Presented at Dakota State University Research Seminar, Madison, SD.

Poster Presentations

- **Ayelazono, B**. (2026, April). *Detection of Neonicotinoids in Eggshells of Wild Bird Species.*
  Presented at South Dakota Academy of Science Annual Symposium, Madison, SD.
- **Ayelazono, B**. (2025, July). *High-Throughput Experimentation and Data Analysis Revealing the Intricate Interplay of Oil Spills and Microplastics Across Diverse Conditions.*
  Presented at the University of Utah Summer Conference. Salt Lake City, UT.
- **Ayelazono, B**. (2025, April). *Survey of Prairie Lakes for Microplastics in Eastern South Dakota.*
  Presented at the National Conference of Undergraduate Research, Pittsburgh, PA.
- **Ayelazono, B.** (2025, March). *Survey of Prairie Lakes for Microplastics in Eastern South Dakota.*
  Presented at Dakota State University Annual Research Symposium. Madison, SD.

Manuscript in Preparation

- **Ayelazono, B**.; Bakker, Kristel. *Detection of Neonicotinoids in Eggshells of Wild Bird Species across South Dakota.*

Published Abstracts

- **Ayelazono, B**.; Shiri, Samira. *High-Throughput Experimentation and Data Analysis Revealing the Intricate Interplay of Oil Spills and Microplastics Across Diverse Conditions. Range* 26:12.
- **Ayelazono, B**. Kristel, Bakker. *Detection of Neonicotinoids in Eggshells of Wild Bird Species across South Dakota.* Proceedings of South Dakota Academy of science: Vol. 105. In press